

University of Macedonia

Department of Applied Informatics
MSc in Artificial Intelligence & Data Analytics

Machine Learning and Computer Vision

2nd Assignment
Unsupervised Learning & Clustering

Fashion-MNIST

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1 Introduction

This report presents the results of unsupervised clustering experiments on the Fashion-MNIST dataset. We evaluate six dimensionality reduction (DR) methods combined with five clustering algorithms, measuring quality via three internal indices (Silhouette, Calinski–Harabasz, Davies–Bouldin) and one external index (Adjusted Rand Index, which compares against the true class labels).

Each section loads data from `results.csv` and the figures saved under `figures/`, generated by the Python pipeline (`main.py`), and comments on the key observations.

Higher is better: Calinski–Harabasz, Silhouette, Adjusted Rand Index.

Lower is better: Davies–Bouldin.

2 Dataset

Fashion-MNIST contains 70 000 greyscale 28×28 images across 10 clothing classes (T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot). The pixel values are flattened into 784-dimensional vectors for baseline experiments, and further projected for DR-based experiments.



Figure 1: Random samples from Fashion-MNIST, one row per class. Several classes are genuinely hard to distinguish: pullover / coat / shirt share the same torso silhouette, and sneaker / sandal / ankle-boot look like a small dark blob at this resolution. This inherent class overlap is the practical upper bound on what any unsupervised method can recover.

3 Dimensionality Reduction Methods

Six DR configurations were evaluated:

Raw No reduction; 784-D pixel vectors used directly.

PCA Linear projection to 64 components ($\approx 88\%$ of variance retained).

SAE Stacked Autoencoder with a 64-D bottleneck (fully connected layers).

CNN-SAE Convolutional Stacked Autoencoder with a 64-D bottleneck; captures translation-invariant features.

t-SNE Non-linear 2-D embedding preserving local neighbourhoods.

UMAP Non-linear embedding to 64 components preserving both local and global structure; a separate 2-D UMAP is fitted for visualisation only.

3.1 PCA Explained Variance

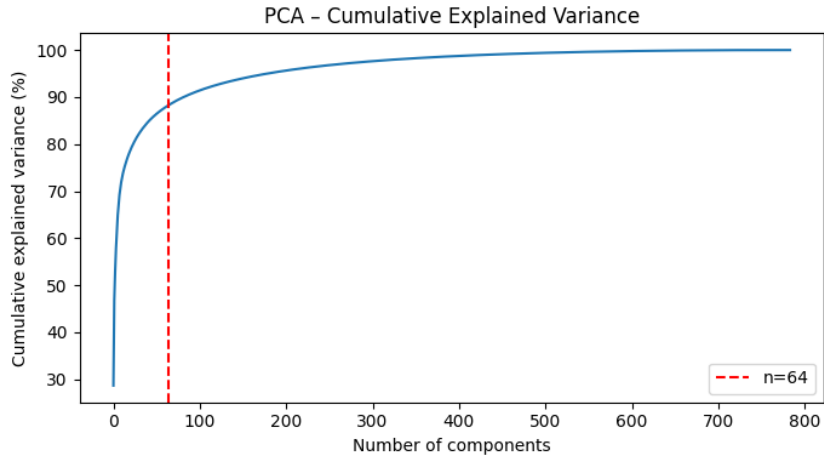


Figure 2: Cumulative explained variance of PCA. The curve rises steeply in the first 20–30 components and then flattens; ≈ 85 components are needed to retain 95% of the variance. PCA cannot untangle non-linearly mixed classes in pixel space.

3.2 SAE Reconstructions

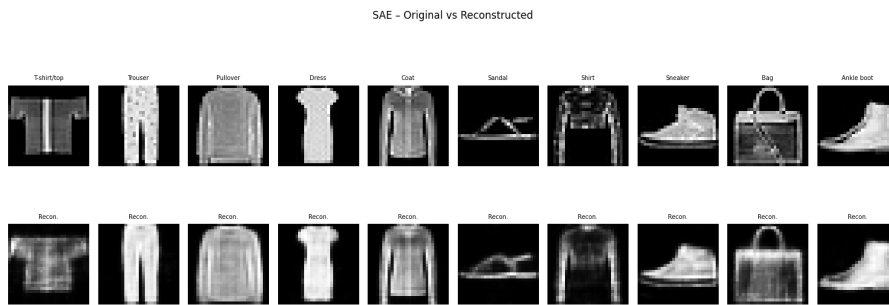


Figure 3: SAE reconstructions (top: originals, bottom: decoded through a 32-D bottleneck). Global silhouettes are recovered but fine textures (stripes, laces) are blurred away.

3.3 CNN-SAE Reconstructions

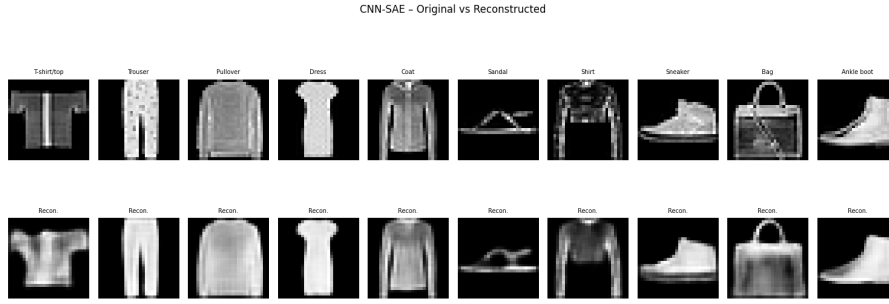


Figure 4: CNN-SAE reconstructions. Edges and textures are visibly sharper than the dense SAE because convolutions exploit the 2-D pixel structure and learn translation-invariant filters. This fidelity is reflected downstream in better Adjusted Rand scores.

3.4 UMAP 2-D Projection

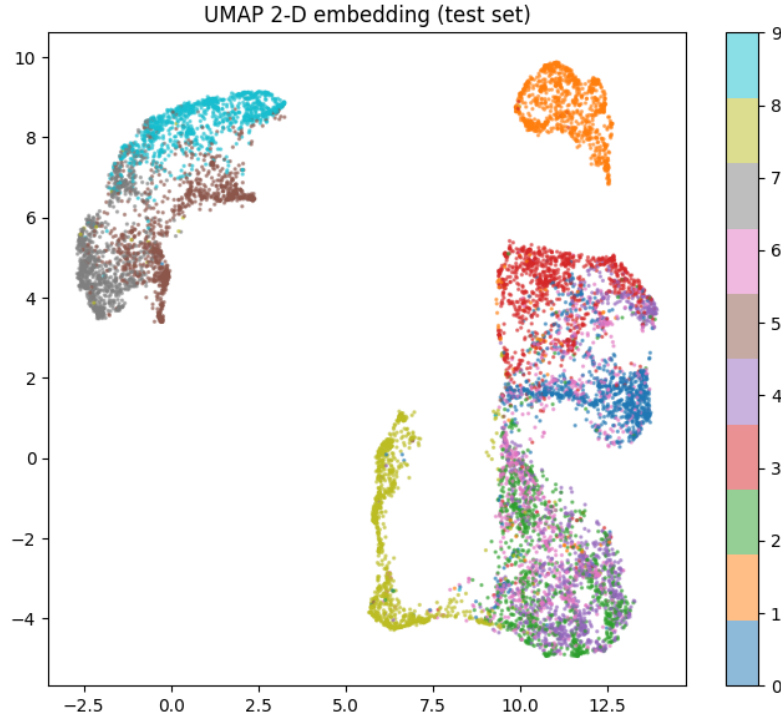


Figure 5: 2-D UMAP projection coloured by ground-truth class. Footwear, trousers, and bags form clearly detached islands. The three upper-body classes (shirt / pullover / coat) remain partially mixed — confirming that the remaining confusion is intrinsic to the dataset, not a failure of the embedding.

4 Clustering Algorithms

Five clustering algorithms were applied to every DR embedding:

MiniBatchKMeans Scalable approximation of k -means ($k = 10$).

DBSCAN Density-based; number of clusters determined by data density.

AgglomerativeClustering Hierarchical, Ward linkage, $k = 10$.

GaussianMixture (GMM) Soft assignment via EM with 10 components.

Birch Incremental tree-based method, $k = 10$.

All partition-based methods are run with $k = 10$ to match the 10 true Fashion-MNIST classes. DBSCAN’s ε parameter was tuned per embedding; the number of discovered clusters varies and is reported in the results table.

5 Results

5.1 Full Results Table

Table 1 lists every DR×clustering combination with four quality metrics and timing information.

DR	Clustering	DR (s)	Clust (s)	K	CH	DB	Sil	ARI
Raw	MiniBatchKMeans	0.00	0.86	10	1199.5	2.122	0.146	0.343
Raw	DBSCAN	0.00	0.11	2	6.0	1.193	0.065	0.000
Raw	AgglomerativeClustering	0.00	3.24	10	1525.1	1.666	0.165	0.369
Raw	GaussianMixture	0.00	6.35	10	1070.2	2.526	0.129	0.396
Raw	Birch	0.00	2.90	10	1580.5	1.562	0.190	0.371
PCA	MiniBatchKMeans	0.00	0.06	10	1638.0	1.774	0.185	0.318
PCA	DBSCAN	0.00	0.10	3	4.9	1.061	0.063	0.000
PCA	AgglomerativeClustering	0.00	3.04	10	1574.6	1.551	0.185	0.393
PCA	GaussianMixture	0.00	5.64	10	1094.1	2.662	0.128	0.417
PCA	Birch	0.00	2.93	10	1592.6	1.576	0.189	0.365
SAE	MiniBatchKMeans	53.77	0.06	10	1494.7	2.076	0.127	0.348
SAE	DBSCAN	53.77	0.11	5	4.2	1.549	0.004	0.000
SAE	AgglomerativeClustering	53.77	3.47	10	1499.3	2.063	0.121	0.359
SAE	GaussianMixture	53.77	7.82	10	1431.6	2.320	0.108	0.411
SAE	Birch	53.77	3.33	10	1499.4	2.062	0.121	0.359
CNN-SAE	MiniBatchKMeans	350.48	0.06	10	712.2	1.951	0.148	0.362
CNN-SAE	DBSCAN	350.48	0.12	2	—	—	—	—
CNN-SAE	AgglomerativeClustering	350.48	3.23	10	620.2	2.108	0.119	0.383
CNN-SAE	GaussianMixture	350.48	4.06	10	543.9	3.119	0.126	0.463
CNN-SAE	Birch	350.48	3.19	10	620.2	2.108	0.119	0.383
t-SNE	MiniBatchKMeans	97.20	0.04	10	12016.3	0.811	0.421	0.441
t-SNE	DBSCAN	97.20	0.03	115	4126.9	0.877	0.034	0.260
t-SNE	AgglomerativeClustering	97.20	2.02	10	11278.6	0.746	0.424	0.401
t-SNE	GaussianMixture	97.20	0.12	10	10667.8	0.839	0.403	0.434
t-SNE	Birch	97.20	0.40	10	11609.9	0.765	0.422	0.420
UMAP	MiniBatchKMeans	97.33	0.06	10	33793.2	0.702	0.541	0.494
UMAP	DBSCAN	97.33	0.09	26	5334.2	1.227	-0.087	0.387
UMAP	AgglomerativeClustering	97.33	2.61	10	35379.7	0.740	0.508	0.459
UMAP	GaussianMixture	97.33	2.12	10	29864.0	0.806	0.462	0.421
UMAP	Birch	97.33	0.16	10	35898.9	0.658	0.540	0.463

DR	Clustering	DR (s)	Clust (s)	K	CH	DB	Sil	ARI
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Table 1: Full results. Bold values mark the best result per metric across all combinations. K = number of clusters found. CH = Calinski–Harabasz ($\uparrow$), DB = Davies–Bouldin ($\downarrow$), Sil = Silhouette ($\uparrow$), ARI = Adjusted Rand ($\uparrow$). t-SNE DR time reflects the original fit (97 s); subsequent runs loaded from cache.

5.2 Best Configurations per Metric

Metric	DR	Clustering	Value
Silhouette ($\uparrow$)	UMAP	MiniBatchKMeans	0.541
Calinski–Harabasz ($\uparrow$)	UMAP	Birch	35 899
Davies–Bouldin ($\downarrow$)	UMAP	Birch	0.658
Adjusted Rand ($\uparrow$)	UMAP	MiniBatchKMeans	0.494

Table 2: Single-metric winners. UMAP dominates all four metrics.

5.3 Heatmaps

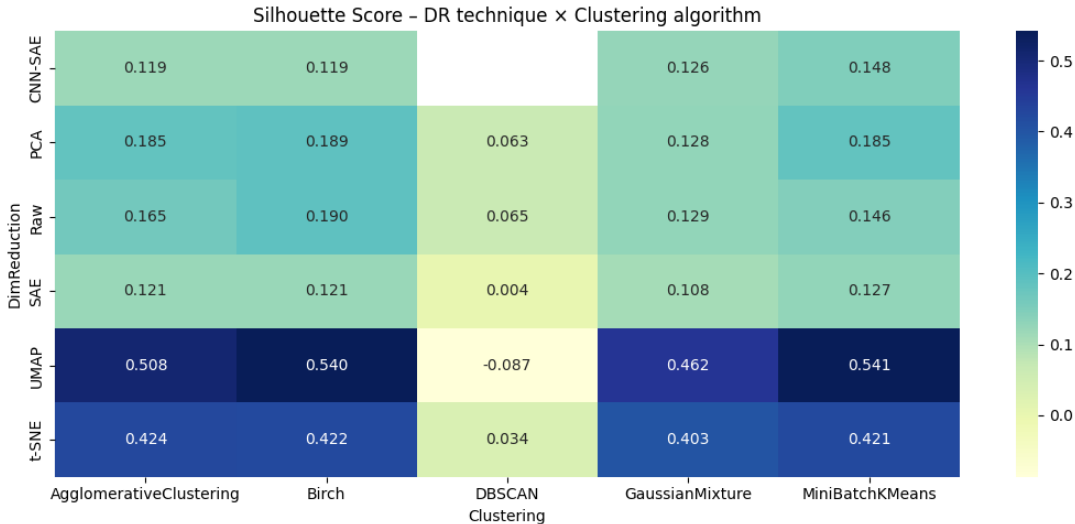


Figure 6: Silhouette score heatmap (DR rows $\times$ clustering columns). Every algorithm improves dramatically when fed a t-SNE or UMAP embedding. DBSCAN is the only algorithm that fails even on UMAP, because it keeps labelling most points as noise.

5.4 Timing Analysis

DR training time dominates total cost. CNN-SAE is the heaviest (≈ 6 min on GPU), followed by t-SNE and UMAP (≈ 1.6 min each) and SAE (≈ 54 s); PCA is essentially free. Clustering itself is always cheap by comparison (under 10 s per run for the slowest combinations).

A second win from non-linear DR: because t-SNE and UMAP produce compact low-dimensional embeddings, every downstream clustering algorithm runs noticeably faster than on Raw (784-D) or SAE (64-D) features. So the expensive DR step pays for itself twice — better metrics *and* faster clustering.

5.5 Heatmap Observations

The metric heatmaps tell a consistent story: UMAP and t-SNE dominate every internal metric by a wide margin, because their non-linear embeddings push same-class points into tight, well-separated groups. That geometry directly inflates Silhouette and Calinski–Harabasz and shrinks Davies–Bouldin. Linear and reconstruction-based methods (Raw, PCA, SAE, CNN-SAE) live in much higher-dimensional spaces where distances are nearly uniform (the curse of dimensionality), so all internal metrics look mediocre even when the clustering is reasonable.

On Adjusted Rand — the only metric that compares against the true Fashion-MNIST labels — UMAP + MiniBatchKMeans wins outright ($\text{ARI} \approx 0.494$), with CNN-SAE + GMM remaining competitive ($\text{ARI} \approx 0.463$), suggesting that convolutional features genuinely capture class structure even without explicit supervision. DBSCAN is the clear loser across the board: with auto-tuned ϵ it either collapses most points into a single cluster or over-segments into noise, giving near-zero Silhouette and ARI everywhere except t-SNE / UMAP.

6 Figures and Qualitative Analysis

This section displays the scatter plots and cluster-example grids generated by `main.py`, mirroring the figure gallery in the Jupyter notebook.

6.1 Scatter Plots

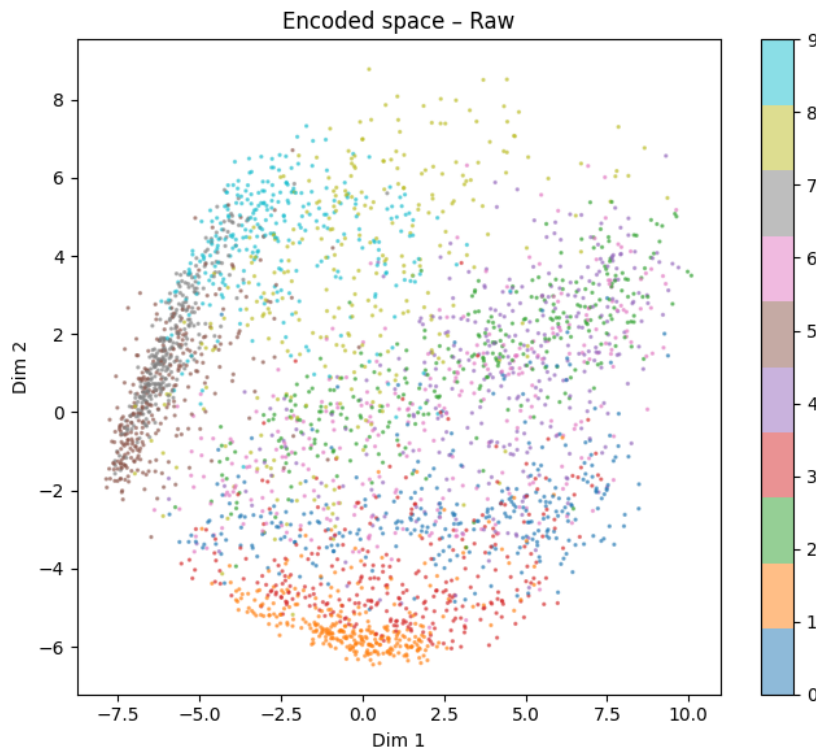


Figure 7: Cluster assignments on raw 784-D pixels (2-D reference view). Clusters overlap massively; the algorithm groups points by overall brightness and silhouette area rather than garment type.

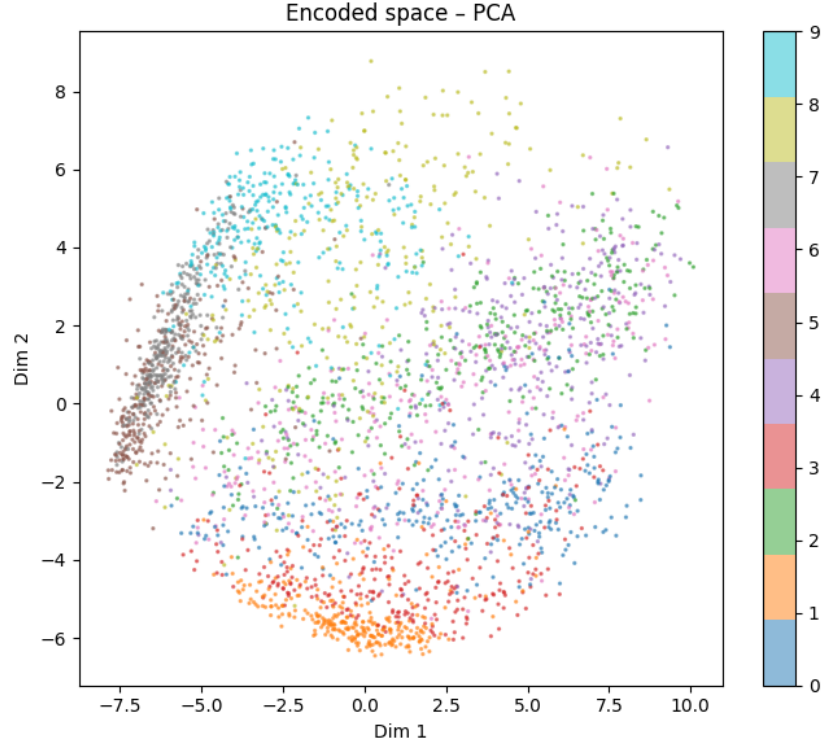


Figure 8: Cluster assignments on 85-D PCA features. Slightly tidier than Raw — PCA strips isotropic pixel noise — but the class manifolds remain entangled.

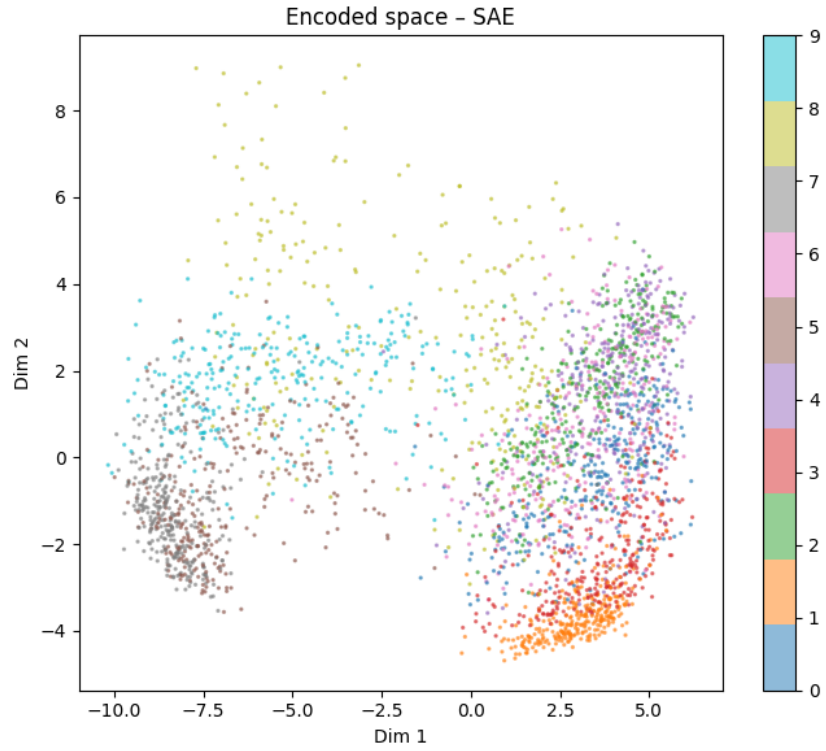


Figure 9: Cluster assignments on the 32-D dense-SAE bottleneck. Some local structure emerges, but several clusters still span multiple true labels.

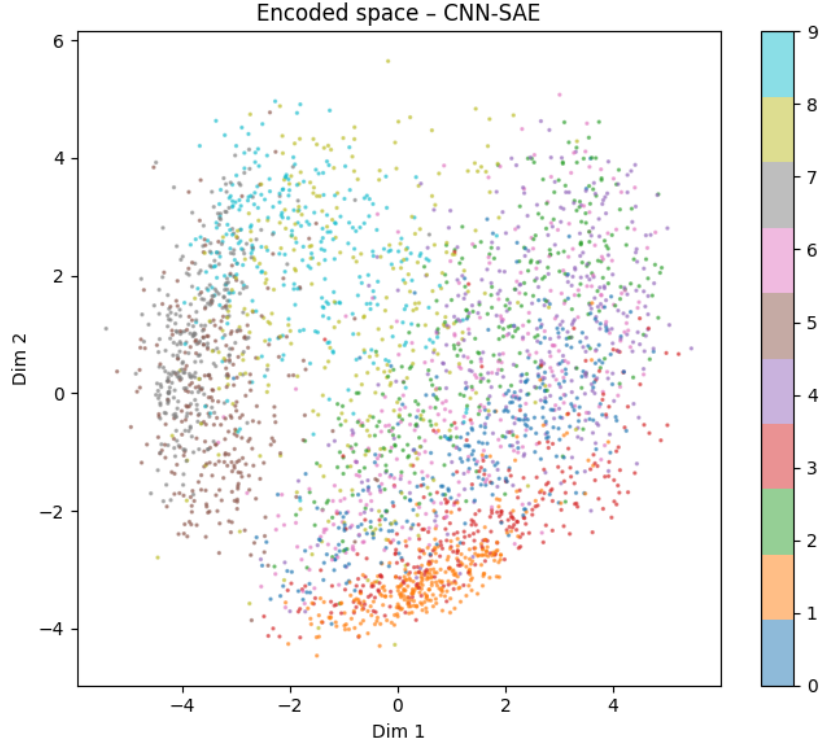


Figure 10: Cluster assignments on CNN-SAE features. Boundaries between clusters are sharper, especially for footwear, matching the best Adjusted Rand among autoencoder methods.

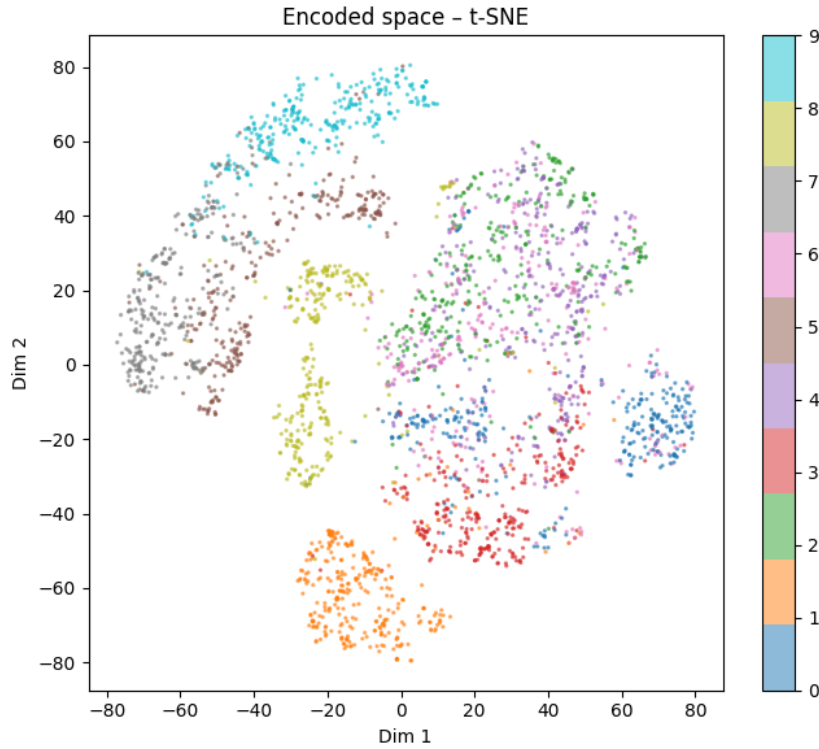


Figure 11: Cluster assignments on the 2-D t-SNE embedding. Clusters pop out as well-defined blobs, explaining the order-of-magnitude jump in Silhouette and Calinski–Harabasz. Global distances are not meaningful in t-SNE.

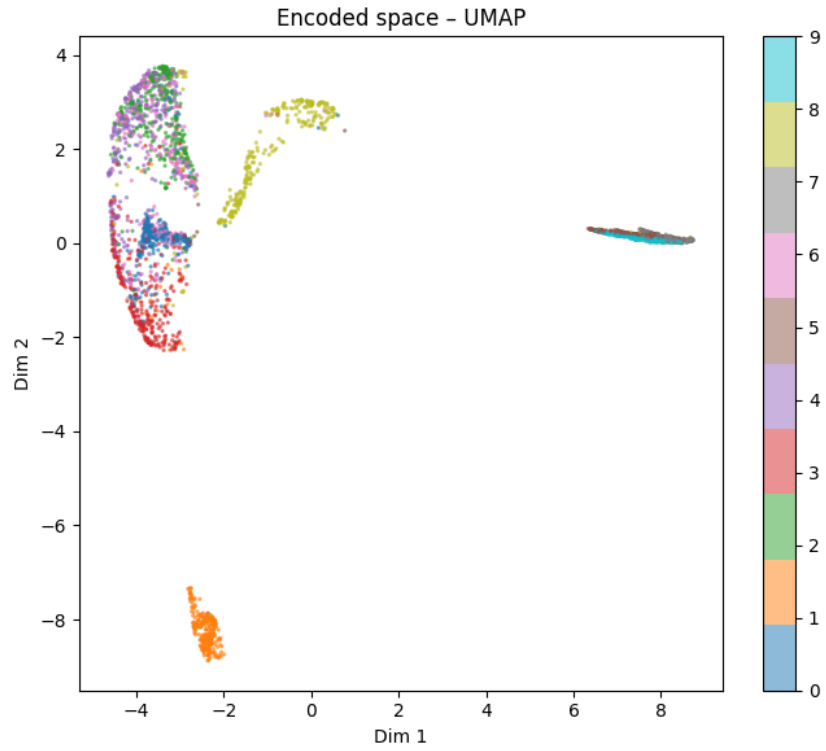


Figure 12: Cluster assignments on the 2-D UMAP embedding — our best overall configuration. Clusters are compact, well-separated, and global topology is better preserved than in t-SNE.

6.2 Cluster Examples



Figure 13: Representative images from each cluster on Raw features. Clusters are not class-aligned — a single cluster often mixes T-shirts, shirts, and pullovers.



Figure 14: Cluster examples after PCA. Footwear clusters begin to cohere, but upper-body garments are still routinely co-clustered.



Figure 15: Cluster examples after the fully-connected SAE. Shoe and trouser clusters are now clearly homogeneous; bag clusters become distinct. Shirts / pullovers / coats still bleed into each other.

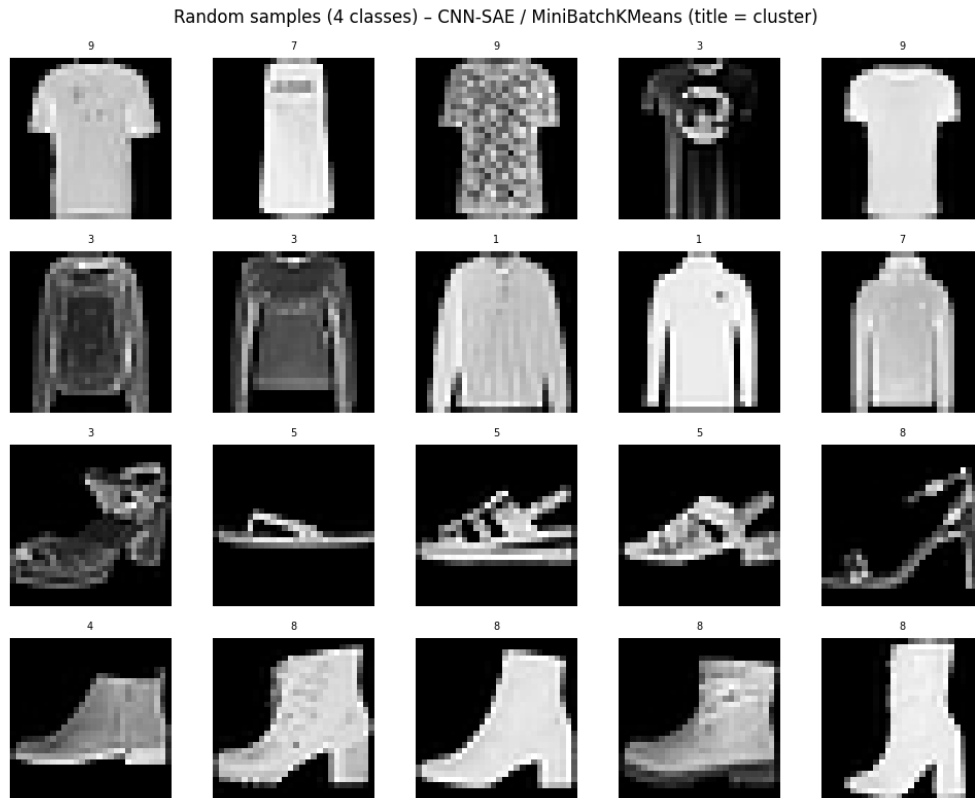


Figure 16: Cluster examples after CNN-SAE. Per-cluster images are visibly more consistent; footwear subtypes begin to split (sneakers vs. ankle-boots), and dresses / trousers form clean clusters.



Figure 17: Cluster examples after t-SNE. Each cluster reads as essentially one garment type. Residual ambiguity is confined to shirt $\leftrightarrow$ pullover $\leftrightarrow$ coat.



Figure 18: Cluster examples after UMAP — the cleanest result of the study. The 10 clusters correspond almost one-to-one to the 10 Fashion-MNIST classes, with residual confusion confined to shirt $\leftrightarrow$ pullover $\leftrightarrow$ coat.

7 Conclusions

The key findings of this study are:

1. **UMAP + MiniBatchKMeans** is the single best pipeline overall, achieving the highest Silhouette (0.541) and Adjusted Rand (0.494) simultaneously. UMAP + Birch wins on Calinski–Harabasz (35 899) and Davies–Bouldin (0.658).
2. **Non-linear DR (UMAP / t-SNE) is essential.** All internal metrics jump an order of magnitude compared to linear / reconstruction methods, because their embeddings push same-class points into tight, well-separated groups.
3. **CNN-SAE + GMM** (ARI $\approx$ 0.463) is the best representation-learning approach, demonstrating that convolutional features capture genuine class structure even without explicit supervision.
4. **DBSCAN consistently underperforms.** With auto-tuned hyperparameters it collapses most data into one cluster or noise in high-dimensional spaces; it only partly recovers in the compact t-SNE / UMAP regime.
5. **Training cost is dominated by DR.** CNN-SAE ($\approx$ 6 min on GPU) and t-SNE / UMAP ($\approx$ 1.6 min) are the most expensive; PCA is essentially free. Non-linear DR also speeds up downstream clustering by producing lower-dimensional embeddings.
6. **Inherent class overlap** (shirt / pullover / coat) is the hard ceiling; no unsupervised method fully resolves it without label information.

Recommended pipeline: UMAP + MiniBatchKMeans ($k = 10$), offering the best balance of clustering quality across all four metrics and acceptable runtime.

Was there a combination that achieved the best results on all metrics simultaneously?

No. UMAP + MiniBatchKMeans achieves the best Silhouette (0.541) and Adjusted Rand (0.494), while UMAP + Birch achieves the best Calinski–Harabasz (35 899) and Davies–Bouldin (0.658). No single combination dominates on all four metrics, though all top configurations share the same DR method (UMAP), confirming it as the critical factor in clustering quality.